

KIRKWALL, United Kingdom

WMO: 030170

Lat: 58.9536N

Lon: 2.9014W

Elev: 16

StdP: 101.14

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.1	-0.1	-5.1	2.5	0.9	-4.1	2.7	1.5	21.2	5.1	19.2	5.7	5.3	260	0.836

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.2	18.2	15.2	17.1	14.4	16.1	13.8	15.9	17.6	15.0	16.4	14.3	15.5	5.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	10.7	16.7	14.3	10.2	15.7	13.7	9.8	15.0	44.6	17.6	42.2	16.5	40.3	15.6	20.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.1	14.8	13.3	DB	-3.1	21.3	1.9	1.4	-4.5	22.3	-5.5	23.1	-6.6	24.0	-7.9	25.0	(4)
(5)			WB	-4.0	17.8	1.8	1.1	-5.3	18.6	-6.3	19.2	-7.3	19.9	-8.6	20.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	4.5	4.4	5.2	6.8	8.7	11.0	13.1	13.2	11.7	9.3	6.8	5.0	(6)	
	DBStd	3.81	1.98	2.20	2.25	2.21	2.13	1.85	1.76	1.65	1.86	2.12	2.33	2.49	(7)	
	HDD10.0	937	172	157	150	97	52	10	1	1	7	37	98	156	(8)	
	HDD18.3	3655	430	390	408	346	299	219	163	159	199	279	347	414	(9)	
	CDD10.0	325	0	0	1	2	11	42	96	99	58	16	1	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
	Wind	WSAvg	6.9	8.0	8.1	7.6	7.0	6.1	6.0	5.4	5.5	6.3	7.2	7.6	7.5	(14)
(15) Precipitation	PrecAvg	1017	105	83	86	57	49	54	56	73	94	112	126	113	(15)	
	PrecMax	1360	178	176	167	114	89	108	107	194	175	236	216	222	(16)	
	PrecMin	745	37	19	25	9	6	9	18	19	33	34	48	41	(17)	
	PrecStd	131	34	38	34	24	20	21	23	34	37	44	41	40	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	10.8	12.1	14.3	17.1	18.3	20.4	20.0	17.7	14.9	12.5	11.3	(19)	
		MCWB	9.1	9.5	9.3	11.2	13.5	15.0	16.8	16.6	14.7	13.2	11.3	10.2	(20)	
	2%	DB	9.1	9.7	10.2	12.3	14.7	16.3	18.4	18.0	16.1	13.6	11.5	10.3	(21)	
		MCWB	8.1	8.7	8.4	9.7	11.5	13.2	15.6	15.1	13.7	12.1	10.5	9.4	(22)	
	5%	DB	8.2	8.7	9.2	11.0	13.2	15.2	17.1	16.8	15.2	12.9	10.9	9.5	(23)	
		MCWB	7.4	7.6	7.8	9.0	10.6	12.7	14.6	14.3	13.3	11.6	10.0	8.6	(24)	
	10%	DB	7.6	7.8	8.5	9.9	12.1	14.1	16.1	15.9	14.4	12.2	10.2	8.6	(25)	
		MCWB	6.7	6.7	7.2	8.2	9.8	12.0	14.0	13.8	12.9	11.1	9.3	7.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.5	10.0	10.0	11.7	13.9	15.4	17.5	17.0	15.5	13.5	11.7	10.5	(27)	
		MCDB	9.9	10.4	11.2	13.6	16.4	17.6	19.6	19.2	16.8	14.4	12.0	11.1	(28)	
	2%	WB	8.3	9.0	8.9	10.2	12.1	14.0	16.1	15.8	14.5	12.6	10.8	9.7	(29)	
		MCDB	8.9	9.6	9.7	11.8	13.9	15.7	17.8	17.3	15.4	13.2	11.3	10.2	(30)	
	5%	WB	7.6	7.8	8.1	9.2	11.1	13.1	15.2	14.9	13.8	12.0	10.1	8.7	(31)	
		MCDB	8.2	8.6	9.0	10.5	12.7	14.6	16.6	16.2	14.6	12.6	10.7	9.3	(32)	
	10%	WB	6.8	6.9	7.3	8.5	10.3	12.3	14.4	14.3	13.2	11.4	9.4	7.9	(33)	
		MCDB	7.5	7.7	8.3	9.6	11.7	13.6	15.6	15.3	14.1	12.1	10.1	8.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.5	4.0	4.6	5.1	5.7	5.3	5.3	5.2	4.8	4.2	3.9	3.8	(35)	
		MCDBR	4.1	4.5	5.5	6.3	7.1	6.7	6.9	6.4	5.6	4.4	4.2	4.4	(36)	
	5% WB	MCWBR	4.0	4.1	4.2	4.2	4.5	4.4	4.4	4.1	3.9	3.8	4.2	4.4	(37)	
		MCWBR	3.9	4.3	5.3	5.7	6.6	6.3	6.4	5.8	5.0	4.1	4.1	4.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.328	0.348	0.366	0.385	0.374	0.371	0.380	0.369	0.360	0.346	0.327	0.309	(40)	
		taud	2.152	2.262	2.310	2.302	2.396	2.442	2.428	2.464	2.469	2.370	2.204	2.098	(41)	
	Ebn at Noon		452	655	767	817	857	865	846	830	775	656	452	335	(42)	
		Edh at Noon	50	73	92	109	106	103	102	92	79	65	47	37	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.32	0.90	2.00	3.53	4.84	4.83	4.50	3.65	2.36	1.19	0.43	0.20	(44)	
	RadStd		0.03	0.09	0.15	0.28	0.33	0.39	0.33	0.26	0.19	0.11	0.04	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.27	N/A	+0.81	N/A	N/A	N/A	-79	N/A	N/A	N/A
Regional (22 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page